

Quick Check

Lesson 4: Germination: Process, Types and Conditions Necessary for Germination in Maize and Bean Seeds

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 4

A group sets up Container A (moist cotton wool) and Container B (dry cotton wool) to test whether maize and bean seeds need water to germinate. Which of these must be kept the SAME in both containers for the test to be fair?

A The type and number of seeds and the temperature

B Whether the seeds are alive or boiled

C The amount of water given

D Whether the cotton wool is wet or dry

In the germination investigation, one container holds boiled and then cooled maize and bean seeds on moist cotton wool with air and room temperature. Why is this container included?

- A** To show that a seed must be alive (viable) to germinate even when water, air and warmth are present
- B** To show that seeds need cold conditions to germinate
- C** To show that seeds germinate faster when they are boiled
- D** To show that seeds need more water than the control container

A learner examines two seedlings. In seedling X, the coleoptile and plumule push up while the cotyledon stays below the soil. In seedling Y, the hypocotyl arches and lifts the cotyledons above the soil. Which statement is correct?

- A** Both X and Y are epigeal
- B** X is bean (epigeal) and Y is maize (hypogeal)
- C** Both X and Y are hypogeal
- D** X is maize (hypogeal) and Y is bean (epigeal)

Maize seeds are fully submerged in boiled, cooled water with a thin layer of cooking oil on top. After 5 days, very few have germinated, even though there is plenty of water. What is the best explanation?

- A** The water was too warm for enzymes to work
- B** The seeds had no soil to grow in
- C** The seeds received too much light
- D** Too little oxygen is available for aerobic respiration, so the seeds cannot release the energy needed for germination

In a germination test, 9 of 10 bean seeds germinated on moist cotton wool at room temperature with no soil. Which conclusion is supported by this evidence?

A Seeds need fertiliser to germinate

B Seeds need soil to germinate

C Seeds germinate only in the dark

D Soil is not necessary for germination if water, oxygen and suitable warmth are present

Which list shows the correct order of events in the germination of a maize or bean seed?

- A** Water uptake, enzymes activated, radicle emerges, plumule emerges
- B** Radicle emerges, water uptake, enzymes activated, plumule emerges
- C** Enzymes activated, plumule emerges, water uptake, radicle emerges
- D** Plumule emerges, radicle emerges, water uptake, enzymes activated

A farmer in Kitale plants healthy-looking maize seeds after good rain. The soil is moist, well aerated and warm, yet some seeds never germinate. Which is the most likely reason?

- A** The seeds are hypogeal, so they cannot germinate
- B** The seeds did not get enough sunlight underground
- C** The seeds are dead (not viable) or dormant
- D** The soil had too much oxygen

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).